A passive notch filter and component tolerance

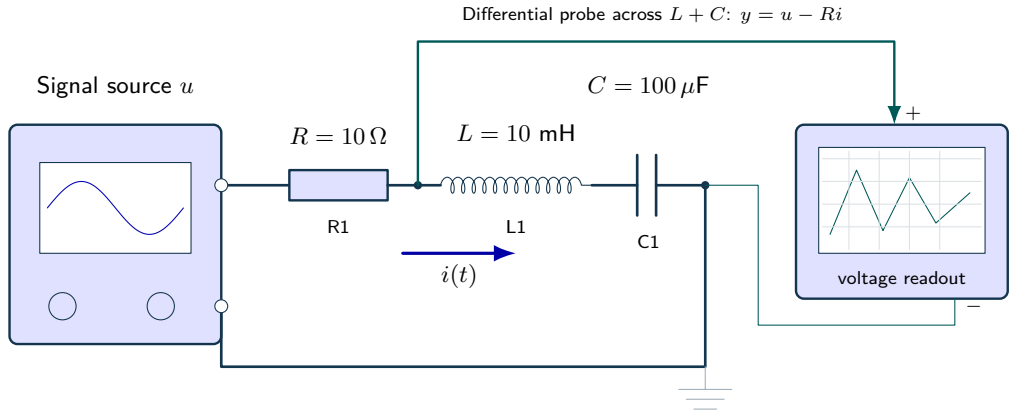
Original learning collection – November 2026

Evidence note. Original teaching scenario. Numerical parameters are deliberately chosen for learning. All generated data and curves are synthetic; no physical measurement is asserted.

1 Design brief

An ideal voltage source drives series R, L and C. Measure the combined voltage $y = v_L + v_C$. Use $L = 10\,\text{mH}$, $C = 100\,\mu\text{F}$ and $R = 10\,\Omega$. Derive the notch transfer, compare it with the capacitor-voltage response, and evaluate the effect of a $\pm 10\%$ capacitance tolerance. Assume ideal components and an unloaded measurement.

2 Physical system



Ideal series network; the output port includes direct feedthrough

Electrical states
Current i [A] and capacitor voltage v_C [V] store magnetic and electric energy.

Circuit equations
$L\dot{i} = u - Ri - v_C$.
$C\dot{v}_C = i$.
Output: $y = u - Ri = v_L + v_C$.

Measurement boundary
The two probe leads span $L + C$. The depicted instrument display is illustrative, not a measured response.

A source excites the series RLC network; an ideal differential probe spans the combined inductor and capacitor. Measuring only the capacitor would produce a different transfer function. Wire placement is schematic.

3 Voltage division

The impedances are $Z_R = R$, $Z_L = sL$ and $Z_C = 1/(sC)$. Measuring across the combined LC section gives

$$G_n(s) = \frac{sL + 1/(sC)}{R + sL + 1/(sC)} = \frac{LCs^2 + 1}{LCs^2 + RCs + 1}.$$

Measuring across C alone instead gives $G_C(s) = 1/(LCs^2 + RCs + 1)$. Both share poles, but their zeros and frequency responses differ.

4 Nominal design

Here $LC = 10^{-6}$ s^2, $RC = 10^{-3}$ s, $\omega_0 = 1/\sqrt{LC} = 1000$ rad/s and $\zeta = (R/2)\sqrt{C/L} = 0.5$. The notch has zeros at $\pm j1000$ and stable poles $-500 \pm j866.0254$. Its DC and high-frequency gains are one.

5 Tolerance analysis

At ideal resonance, the inductor and capacitor voltages cancel in their sum. A capacitance factor a moves the notch to $\omega_0/\sqrt{a}$; a 10 percent tolerance is not a symmetric 10 percent frequency change. The first-order sensitivity is $\delta\omega_0/\omega_0 \simeq -\delta C/(2C)$.

6 Physical interpretation

A zero in the measured sum does not mean zero branch current or zero energy in the individual elements. Resistance in the inductor, source impedance and output loading reduce notch depth and alter the ideal transfer function. The logarithmic plot is clipped only for display at very small magnitudes.

7 What to learn from the experiment

Output selection changes zeros even when the physical circuit is unchanged. An ideal notch is sensitive to component tolerance and omitted losses.

8 System sketch

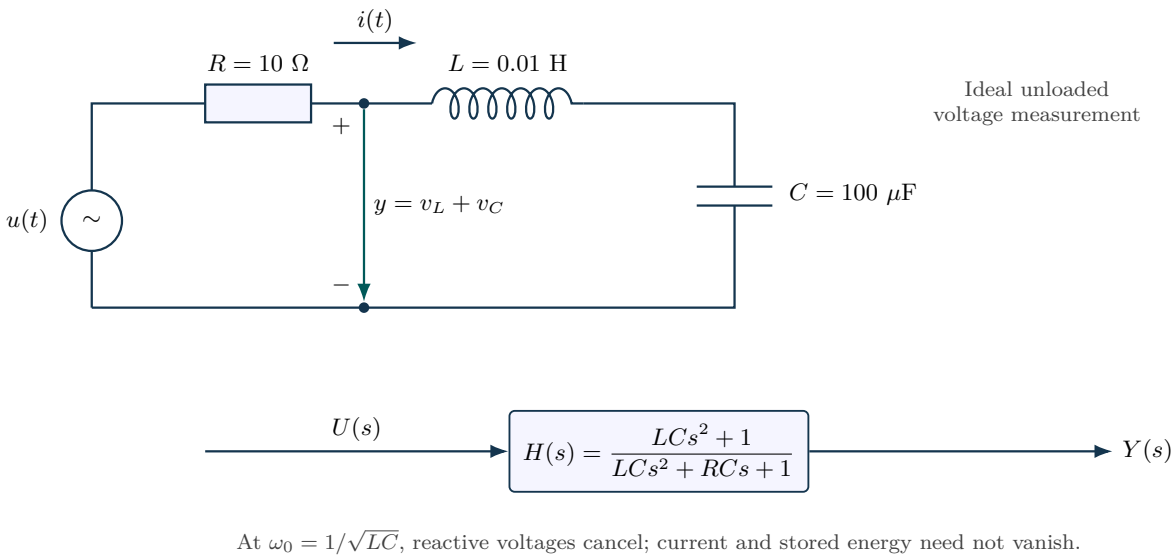
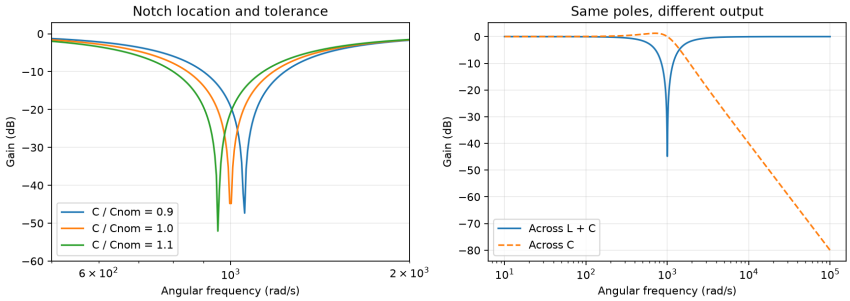


Figure 1. Series RLC network with output measured across L and C together. The output probe spans those two components, not the whole source.

9 Executed computational evidence

```
C factor=0.90: notch=1054.092553 rad/s (167.764040 Hz)
C factor=1.00: notch=1000.000000 rad/s (159.154943 Hz)
C factor=1.10: notch=953.462589 rad/s (151.748284 Hz)
```



Method references. The equations use standard balance laws and feedback-control methods. Further explanation: `https://apmonitor.com/pdc/` and `https://fbsbook.org/`. Implementation documentation: `https://python-control.readthedocs.io/en/0.10.2/`. See the repository reference catalog for individual topics and videos.